230-1 Worksheet: Directional derivatives

November 12, 2024

1. The temperature at a point (x, y, z) is given by the function $T(x, y, z) = \sqrt{2y + xz}$. A bug is flying around and is currently at position $(1, 1, 0)$
   (a) Determine the instantaneous rate of change of temperature if the bug flies towards the origin.
   (b) In which direction should the bug fly to have the temperature decrease fastest from $(1, 1, 0)$.
   (c) Give a direction in which the bug could fly to maintain its current temperature if it is still at $(1, 1, 0)$.

(a) Keyword: rate of change Need: $D_{\vec{u}}T(1,1,0)$.

$$\vec{\nabla}T = \left\langle \frac{\partial T}{\partial x}, \frac{\partial T}{\partial y}, \frac{\partial T}{\partial z} \right\rangle$$

$$\vec{\nabla}T = \left\langle \frac{z}{2\sqrt{2y+xz}}, \frac{1}{\sqrt{2y+xz}}, \frac{x}{2\sqrt{2y+xz}} \right\rangle.$$

$$\vec{\nabla}T(1,1,0) = \left\langle 0, \frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}} \right\rangle$$

$$\vec{v} = \langle 0,0,0 \rangle - \langle 1,1,0 \rangle = \langle -1,-1,0 \rangle.$$

$$\vec{u} = \frac{\vec{v}}{|\vec{v}|} = \left\langle \frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0 \right\rangle.$$

$$D_{\vec{u}}T(1,1,0) = \vec{\nabla}T(1,1,0) \cdot \vec{u}$$

$$= \left\langle 0, \frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}} \right\rangle \cdot \left\langle \frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0 \right\rangle.$$

$$= \frac{1}{\sqrt{2}} \cdot -\frac{1}{\sqrt{2}} = -\frac{1}{2}.$$

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   (a) Determine the instantaneous rate of change of temperature if the bug flies towards the origin.
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(b) Key word: decrease fastest
Need: $-\nabla T(1,1,0)$

$$-\nabla T(1,1,0) = -\langle 0, \frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}} \rangle = \langle 0, -\frac{1}{\sqrt{2}}, -\frac{1}{2\sqrt{2}} \rangle$$

(c) Key word: maintain
Need: A direction $\vec{u}$ such that $D_{\vec{u}} T(1,1,0) = 0$

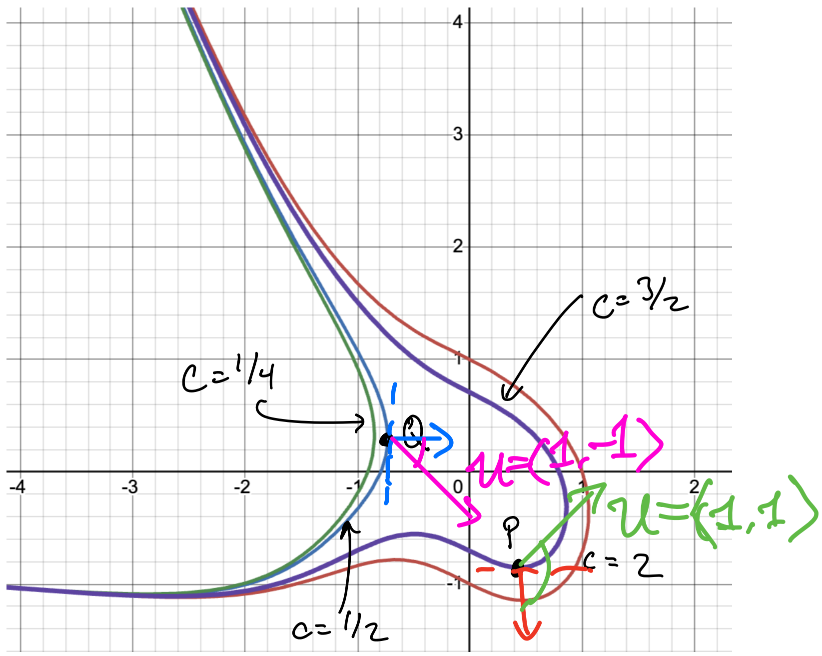
i.e. $\vec{u} \cdot \overrightarrow{\nabla T}(1,1,0) = 0$

Say $\vec{u} = \langle a, b, c \rangle$, then $\langle a, b, c \rangle \cdot \langle 0, \frac{1}{\sqrt{2}}, -\frac{1}{2\sqrt{2}} \rangle = 0$

i.e. $\frac{1}{\sqrt{2}} b - \frac{1}{2\sqrt{2}} c = 0$

Any $\vec{u} = \langle a, b, c \rangle$ such that $b = \frac{1}{2} c$ satisfies.
for example, $\vec{u} = \langle 1, \frac{1}{2}, 1 \rangle$.

2. The level curves of a function $f(x, y)$ are shown below



(a) At the point P sketch what the $\nabla f(P)$ should look like.

(b) Is $D_{\mathbf{u}}f(P)$ positive, negative, or 0 if $\mathbf{u}$ is in the direction of $\langle 1, 1\rangle$.

(c) At the point Q sketch what the $\nabla f(P)$ should look like.

(d) Is $D_{\mathbf{u}}f(Q)$ positive, negative, or 0 if $\mathbf{u}$ is in the direction of $\langle 1, -1\rangle$.

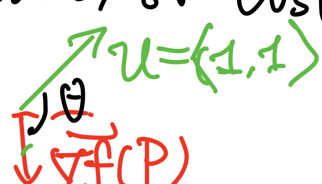
(a) The red arrow (pick the normal vector such that c increases).

(b) $D_u f(P) = \vec{\nabla f} \cdot \frac{\vec{u}}{|\vec{u}|} = |\vec{\nabla f}| \cdot \cancel{|\vec{u}|} \cdot \cos\theta \cdot \frac{1}{\cancel{|\vec{u}|}}$

θ is the angle between $\vec{\nabla f}(P)$ and $\vec{u} = \langle 1, 1\rangle$

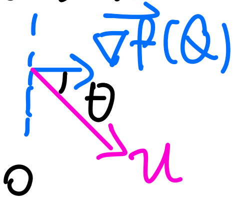
which is obtuse, so $\cos\theta < 0$. $\Rightarrow D_u f(P) < 0$

$\Rightarrow$ negative



(c) The blue arrow (pick the normal vector such that c increases).

(d) Similar as in (b). θ is the angle between $\vec{\nabla f}(Q)$
$\vec{u}=\langle 1,-1\rangle$, which is acute,

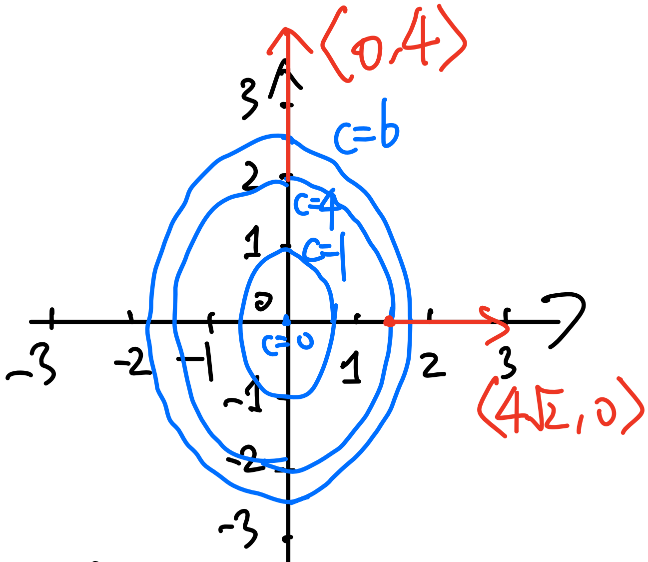


so $\cos\theta<0 \Rightarrow D_u f(Q)>0$
$\Rightarrow$ Positive

3. For the function $f(x, y) = 2x^2 + y^2$ complete the following tasks.

   (a) Sketch a contour graph of $f(x, y)$ consisting of the level curves of f at $c = 0, 1, 4, 6$
   (b) What are $\nabla f(\sqrt{2}, 0)$? and $\nabla f(0, 2)$? Sketch these on your graph.
   (c) In what direction from the point $(0, 2)$ does f increase most rapidly? What is the rate of increase? (i.e. what is $D_{\mathbf{u}}f(0, 2)$ in this direction).

(a)



(b) $\nabla f=\langle 4x,2y\rangle$

$\nabla f(\sqrt{2},0)=\langle 4\sqrt{2},0\rangle$

$\nabla f(0,2)=\langle 0,4\rangle$

(c) Key word: increase most rapidly

In the direction $\vec{v} \overset{def}{=} \vec{\nabla f}(0,2)=\langle 0, 4\rangle$

$\vec{u}=\frac{\vec{v}}{|\vec{v}|}=\langle 0,1\rangle$

Rate of increase $D_{\vec{u}}f(0,2)=\langle 0,4\rangle\cdot\langle 0,1\rangle=4.$